SubD topology patterns

42 solved problems for subdivision modelling. Quads only into the subdivision; poles on flat areas; support loops with Bevel, Shape 1, 2 segments. Every pattern is a real, checked mesh.

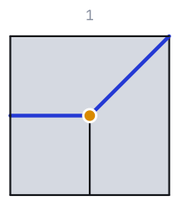
● 3 edges ● 5 edges ● 6 or more

■ Shaded: the part that curves

Renders: subdivided, cage in blue

REDUCING LOOPS

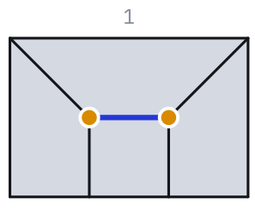
DO 2 → 1



Two rows become one. The extra loop turns the corner and leaves at the side, so it needs room there.

REDUCING LOOPS

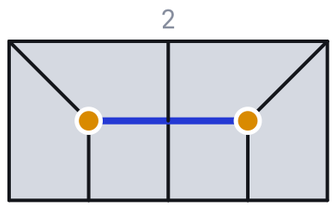
DO 3 → 1



Three in, one out. The middle loop ends between two 3-edge poles.

REDUCING LOOPS

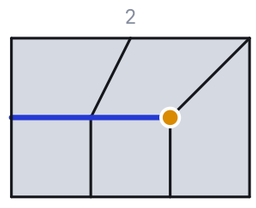
DO 4 → 2



Halves a whole row. The workhorse for changing density.

REDUCING LOOPS

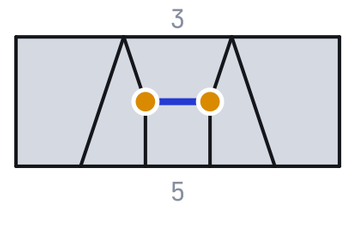
DO 3 → 2



An odd step: one loop comes in from the side and ends at a 3-edge pole.

REDUCING LOOPS

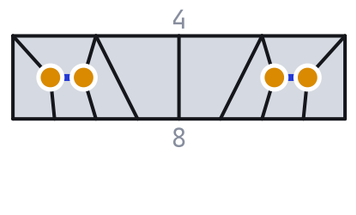
DO 5 → 3



A 3 → 1 in the middle, plain columns on both sides.

REDUCING LOOPS

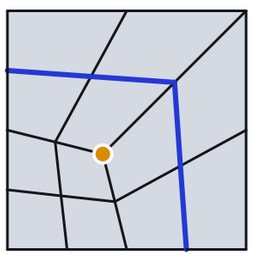
DO 8 → 4



Two 3 → 1 with plain columns between. Two 4 → 2 side by side would leave a 6-edge pole where they meet.

POLES

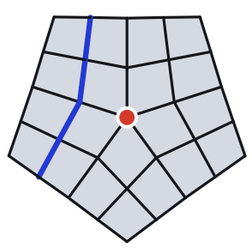
DO N-pole turns a loop



Three patches meet at a 3-edge pole. The blue loop comes in at the bottom and leaves at the side.

POLES

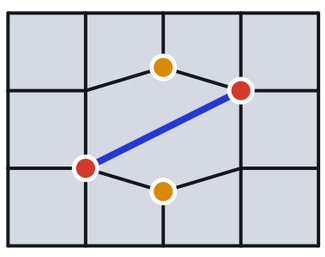
DO E-pole



Five patches meet at a 5-edge pole; the blue loop bends round it. Typical where a form branches off.

POLES

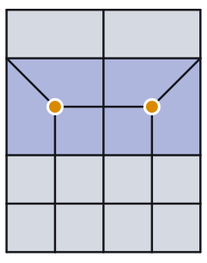
DO Poles come in pairs



Edge → Rotate Edge CW on the blue edge: two 3s and two 5s. A 3 and a 5 cancel out, so the grid around stays regular.

POLES

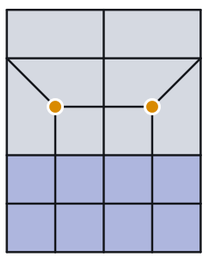
AVOID Pole on the curve



The shaded band bends. A reduction there puts its poles right in the highlight.

POLES

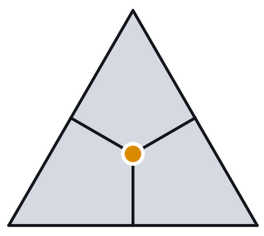
DO Pole on the flat



Same patch: the dense rows carry the curve, and the reduction sits on the flat part.

SHAPES INTO QUADS

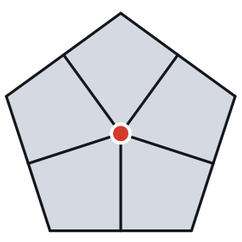
DO Triangle → 3 quads



A point in the middle and one on each edge. The middle becomes a 3-edge pole.

SHAPES INTO QUADS

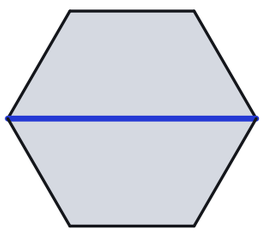
DO Pentagon → 5 quads



The same trick for 5 sides; the middle becomes a 5-edge pole.

SHAPES INTO QUADS

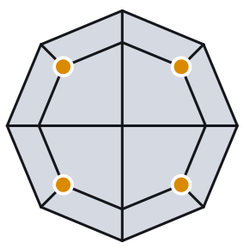
DO 6 sides → 2 quads



One cut across with Connect Vertex Path (J).

SHAPES INTO QUADS

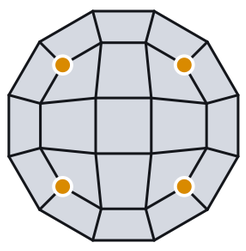
DO Grid Fill, 8 sides



A 2×2 grid. Face → Grid Fill. Four 3-edge poles land on the rim.

SHAPES INTO QUADS

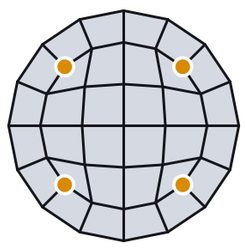
DO Grid Fill, 12 sides



A 3×3 grid, the same four 3-edge poles.

SHAPES INTO QUADS

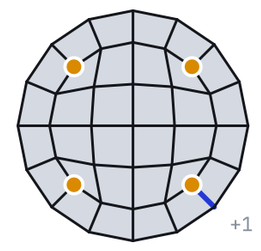
DO Grid Fill, 16 sides



A 4×4 grid. The cap for a 16-sided cylinder.

SHAPES INTO QUADS

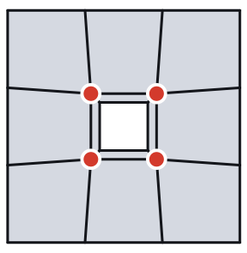
DO Odd edge count



15 edges can't be filled with quads. Add one cut (blue) through the whole ring for 16, then Grid Fill.

CIRCLES

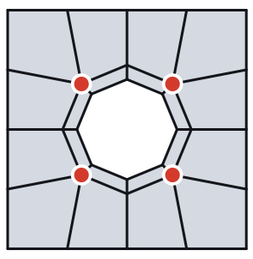
AVOID 4 segments



One face can only give a square. It subdivides to a soft square, not a circle.

CIRCLES

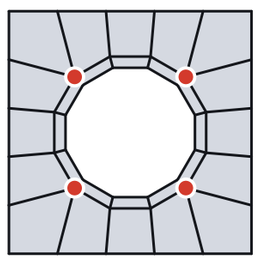
DO 8 segments



From a 2×2 block: the smallest hole that reads as round. The block's corners become 5-edge poles; keep them on the flat.

CIRCLES

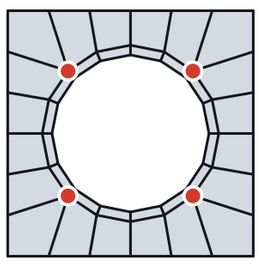
DO 12 segments



From a 3×3 block.

CIRCLES

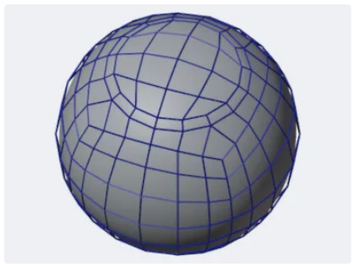
DO 16 segments



From a 4×4 block. Round enough for most props.

CIRCLES

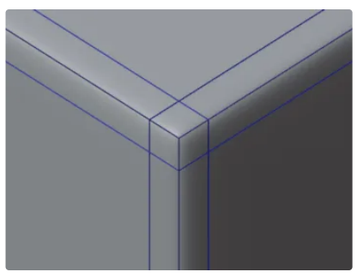
DO Circle on a curve



A 4×4 block on a cube made round, then the whole cage pushed onto a sphere.

CORNERS

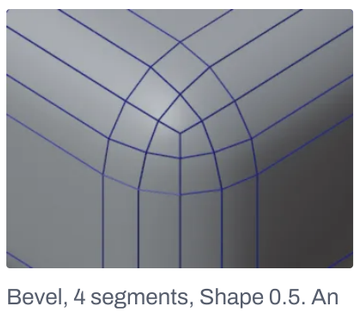
DO Support-loop corner



Bevel, Shape 1, 2 segments. The three loops meet in a 3-edge pole right on the corner.

CORNERS

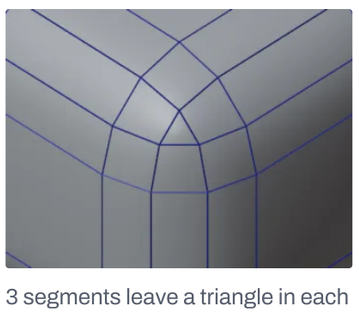
DO Rounded corner



Bevel, 4 segments, Shape 0.5. An even number of segments fills the corner with quads around one 3-edge pole.

CORNERS

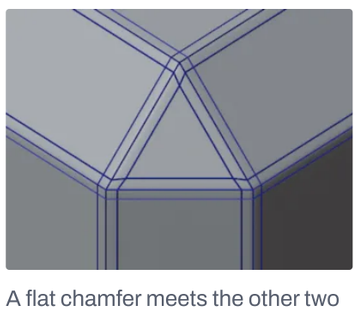
AVOID Odd segments



3 segments leave a triangle in each corner: a pinch in the subdivision.

CORNERS

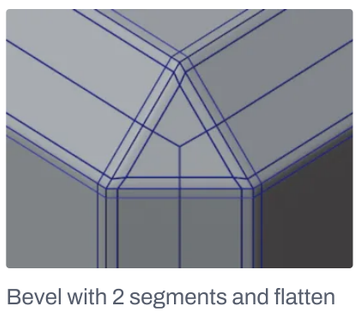
AVOID Chamfer, 1 segment



A flat chamfer meets the other two in a triangle.

CORNERS

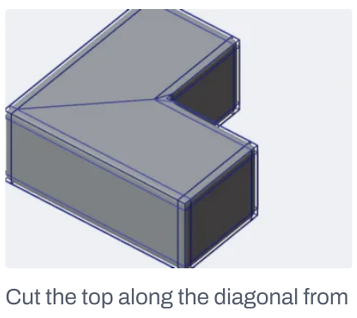
DO Chamfer in quads



Bevel with 2 segments and flatten the middle loop: the corner is three quads. Then support loops.

CORNERS

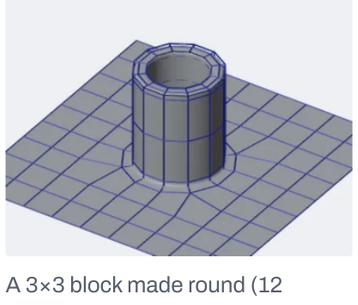
DO Inside corner



Cut the top along the diagonal from the inside corner before you bevel. All quads.

CYLINDERS

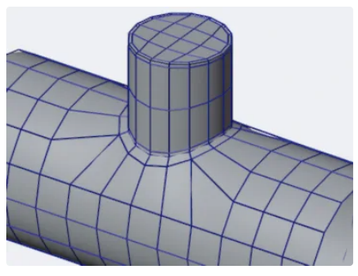
DO Pipe on a wall



A 3×3 block made round (12 sides), Inset, then Extrude. Loops at the base and the lip.

CYLINDERS

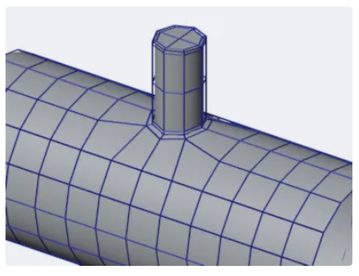
DO T-joint, same count



A 4×4 block of a 16-sided pipe gives the 16 sides of the branch.

CYLINDERS

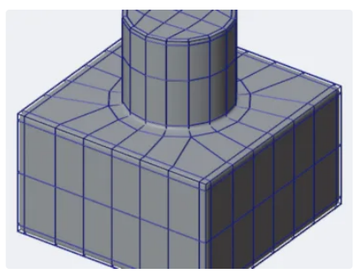
DO Branch, fewer sides



A 2×2 block gives an 8-sided branch. The pipe's loops stay untouched.

CYLINDERS

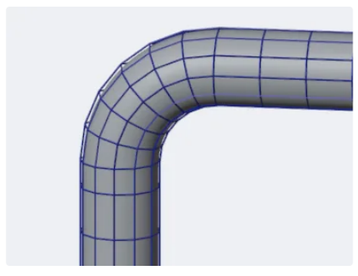
DO Cylinder on a box



The box top is gridded so its middle 4×4 block can become the circle.

CYLINDERS

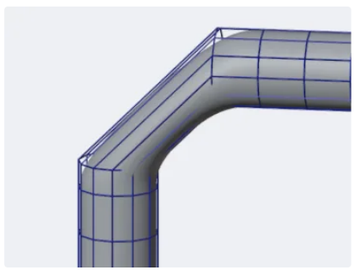
DO Elbow



Rings every 15° through the bend, the same spacing as on the straight parts.

CYLINDERS

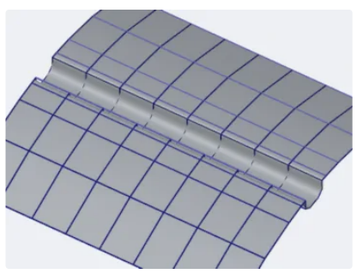
AVOID Elbow, too few rings



No rings in the bend: the inside pinches and the outside goes flat.

SURFACE DETAILS

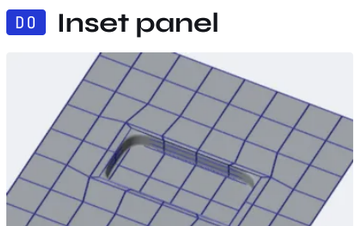
DO Panel line



A groove is a strip of loops: support, edge, wall, floor, and the same back up.

SURFACE DETAILS

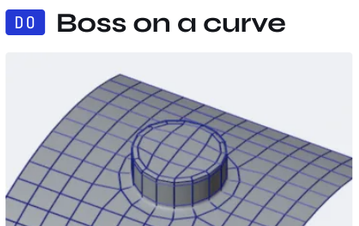
DO Inset panel



Inset (I) the faces, Inset again for the support, then move them down.

SURFACE DETAILS

DO Boss on a curve



The circle comes from the grid; extrude it and add loops at the base and the top.

SURFACE DETAILS

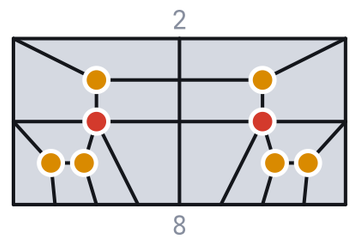
DO Hole in a pipe



The same 4×4 block, deleted. A short wall down gives the hole its thickness.

DENSITY AND FLOW

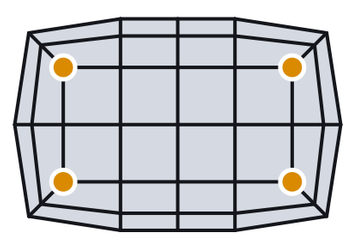
DO 8 → 4 → 2



Stack reductions to go from dense to light over a flat area. Each row adds its poles; keep them off the curves.

DENSITY AND FLOW

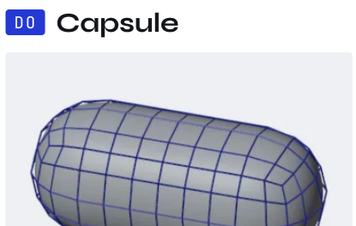
DO Rounded rectangle



Loops follow the outline. The four 3-edge poles sit at the inner corners, on the flat.

DENSITY AND FLOW

DO Capsule



A gridded box with its ends pushed round: all quads, the poles on the ends.